

1

15.240 cm

6.868 cm

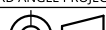
10.597 cm

6.785 cm

9.986 cm

2

1

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .XX = ±.0- ANGULAR = ± ° .XXX = ±.00- FRACTIONAL = ± .XXXX = ±.000- SURFACE FINISH 	NAME		DATE		TITLE	
	DRAWN		06/28/2026			
	CHECKED					
	APPROVED					
DO NOT SCALE DRAWING					SIZE A DWG NO. REV.	
BREAK ALL SHARP EDGES AND REMOVE BURRS						
THIRD ANGLE PROJECTION						
	MATERIAL		FINISH		SCALE 1:2 WEIGHT SHEET 1 of 1	

Workplace area

X coord

Max reach from origin:

$$17.78 + 15.24 + 6.97 \approx 40 \text{ cm}$$

Picking zone around 60% - 80% of max

$$0.6 \times 40 \text{ cm} = 24 \text{ cm}$$

$$0.8 \times 40 \text{ cm} = 32 \text{ cm}$$

$x = 24$ to 32 cm, take the middle at 28 cm

y coord

0 cm

Z coord

tomato center height

Shoulder is 10.6 cm above table



reasonably assume tomato center around 15 - 20 cm above table
take the middle at 18

$$18 - 10.6 = 7.4 \text{ cm}$$

aim point $(28, 0, 7.4 \text{ cm})$

Relative to: 28 cm forwards table
 18 cm above table

Robot coords: $(28, 0, 7.4)$

camera mount location

Let the camera pitch be 45°

Let the camera height be 65 cm

Let mount sit b cm behind origin

$$\text{cam}_x = -b$$

$$\text{target}_x = 28$$

$$\begin{aligned} \text{horiz_dist} &= \text{target}_x - \text{cam}_x \\ &= 28 + b \end{aligned}$$

$$\text{target_height} = 18 \text{ cm}$$

$$\begin{aligned} \text{vert_dist} &= \text{cam_height} - \text{target_height} \\ &= 65 - 18 \\ &= 47 \text{ cm} \end{aligned}$$

$$\text{Pitch} = \tan^{-1}(\text{vert_dist} / \text{horiz_dist})$$

$$45 = \tan^{-1}(47 / (28 + b))$$

$$\tan 45 = 47 / (28 + b)$$

$$1 = \frac{47}{28 + b}$$

$$b = 19$$

I don't like 19 , so b is 20 cm

stereo depth equation

$$Z = FB / d$$

F - focal length
 B - baseline (dist between center)
 d - disparity

Big disparity $\rightarrow$ far
small " $\rightarrow$ close

what happens when disparity is slightly off

suppose disparity has 1px error

$$\frac{dz}{dd} = \frac{-fB}{d^2}$$

Im ignoring the neg

$$Z = FB / d$$
$$d = FB / Z$$

$$dz = \frac{fB}{d^2} \cdot dd$$

$$dz = \frac{fB}{\frac{fB^2}{Z^2}} \cdot dd$$

$$= \cancel{fB} \cdot \frac{Z^2}{\cancel{fB^2}} \cdot dd$$

$$= \frac{Z^2}{fB} \cdot dd$$

so basically as Z increases, depth error is worse

Gets better when f is larger because cam has

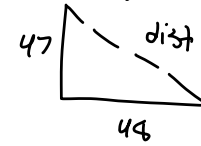
better resolution

with mount 20 cm behind origin

$$\text{horiz_dist} = 48 \text{ cm}$$

$$\text{vert_dist} = 47 \text{ cm}$$

$$\text{dist} = \sqrt{48^2 + 47^2}$$
$$\approx 67$$



$$dz = 0.67^2 / (950 \cdot 0.1) \cdot 1$$
$$= 0.004725$$

4.73mm per 1px disparity error

I think that's fine low